Amrita School of Artificial Intelligence, Faridabad
Amrita Vishwa Vidyapeetham

Course Code: **23MAT214**

Mathematics for Intelligent System-4

Part- 1.6

Shift Invariant Filters

BTech 4th Sem

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System

Input	System	Output
sound	echo effect	echoed sound
image	blur filter	blurred image
stock prices	smoothing	trend

A **system** takes an input signal and produces an output signal: $y = H(x)$

Linearity:

$$H(ax_1 + bx_2) = aH(x_1) + bH(x_2)$$

Examples:

- electrical circuits
- sound reflections
- digital filters

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The Shift Operation

We now define a very important operation: **shifting a signal**.

Define shift operator S:

$$(Sx)[n] = x[n-1]$$

Example

$$x = (4, 2, 7, 1)$$

Shifted signal:

$$Sx = (0, 4, 2, 7)$$

We inserted zero at the beginning.

The signal moved right.

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SHIFT-INVARIANT FILTERS

Definition

System H is shift-invariant if:

$$H(Sx) = S(Hx)$$

$where,\ S$ is the **shift operator**.

Example:

$$x = (3, 5, 2, 1)$$

Then

$$Sx = (0, 3, 5, 2)$$

Real-Life Case

Example: echo in a room.

- If a clap produces an echo after 0.5 seconds.

Then:

- clap now → echo later
- clap 2 seconds later → echo also 2 seconds later.

The system does not change over time.

Meaning:

- If the input is delayed by one unit, the output is also delayed by one unit.
- The system behavior does not depend on the absolute time.

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Basis Signals

Define vectors

$$e_0 = \begin{bmatrix}1\\0\\0\\0\end{bmatrix}$$

$$e_1 = \begin{bmatrix}0\\1\\0\\0\end{bmatrix}$$

$$e_2 = \begin{bmatrix}0\\0\\1\\0\end{bmatrix}, \ldots\ldots\ldots$$

Each represents a **single spike (impulse)**.

Any Signal Can Be Written Using These

Example signal

$$x = (2, 3, 1, 4)$$

We can write

$$x = 2e_0 + 3e_1 + 1e_2 + 4e_3$$

Check:

$$2e_0 = \begin{bmatrix}2\\0\\0\\0\end{bmatrix}$$

$$3e_1 = \begin{bmatrix}0\\3\\0\\0\end{bmatrix}$$

Add them:

$$\begin{bmatrix}2\\3\\1\\4\end{bmatrix}$$

So every signal is a sum of impulses.

Recall

We have a system H.

Input signal: x
Output signal: $y = H(x)$

S is the **shift operator**.

Example:

$$x = (3, 5, 2, 1)$$

Then

$$Sx = (0, 3, 5, 2)$$

Each impulse is simply a **shifted version of the first impulse**.
Observe:

$$e_1 = Se_0$$

because shifting e_0

$$e_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

gives

$$Se_0 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

which equals e_1.
Similarly

$$e_2 = S^2 e_0$$

Similarly: $e_k = S^k e_0$

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Response of System to Impulse

Define

$$h = H(e_0)$$

This vector is called **impulse response**.

Example

$$h = \begin{bmatrix} 1 \\ 0.5 \\ 0.25 \end{bmatrix}$$

Meaning:

Input spike produces

time	output
0	1
1	0.5
2	0.25

This models echo.

Now Compute the Response to e_1

We know

$$e_1 = Se_0$$

Apply the system:

$$H(e_1) = H(Se_0)$$

Now apply **shift invariance**:

$$H(Sx) = S(Hx)$$

so

$$H(Se_0) = S\big(H(e_0)\big)$$

But

$$H(e_0) = h$$

Thus

$$H(e_1) = Sh$$

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$$\boldsymbol{H(e_1) = Sh}$$

Meaning of Sh

Let

$$H(e_0) = h = \begin{bmatrix} 1 \\ 0.5 \\ 0.25 \end{bmatrix}$$

Shift it:

$$\boldsymbol{H(e_1)} = Sh = \begin{bmatrix} 0 \\ 1 \\ 0.5 \\ 0.25 \end{bmatrix}$$

So the entire impulse response moves one step to the right.

$$H(e_2) = S^2 h = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0.5 \\ 0.25 \end{bmatrix}$$

Therefore, $\boldsymbol{H(e_k) = S^k h}$

Response to e_2

Now notice

$$e_2 = S^2 e_0$$

Apply system:

$$H(e_2) = H(S^2 e_0)$$

Using shift invariance again

$$H(S^2 x) = S^2 H(x)$$

$$H(S^2 e_0) = S^2 H(e_0)$$

So

$$H(e_2) = S^2 h$$

$h = H(e_0)$

Real-Life Interpretation: Imagine a room echo system.

Impulse at time 0:

clap
↓
echo
pattern
1
0.5
0.25

Impulse at time 1:

(clap delayed)
↓
echo pattern also delayed
0
1
0.5
0.25

Impulse at time 2:

echo pattern
0
0
1
0.5
0.25

So the **same echo pattern just shifts in time**.

That is exactly what shift invariance means.

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Decompose Any Signal Using Basis Vectors

Let e_k be standard basis:

$$e_0 = (1, 0, 0 \dots)^T,$$
$$e_1 = (0, 1, 0 \dots)^T,$$

Any signal:

$$x = \sum_{k=0}^{N-1} x[k] e_k$$

We are working in the vector space:

$$\mathbb{R}^N \; or \; \mathbb{C}^N$$

The **standard basis vectors** are:

$$e_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \end{bmatrix} \quad e_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \end{bmatrix} \quad e_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ \vdots \end{bmatrix}$$

Suppose: $x = \begin{bmatrix} 2 \\ 5 \\ 3 \\ 7 \end{bmatrix}$

Now look carefully:

$$2e_0 = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix}, 5e_1 = \begin{bmatrix} 0 \\ 5 \\ 0 \\ 0 \end{bmatrix}, 3e_2 = \begin{bmatrix} 0 \\ 0 \\ 3 \\ 0 \end{bmatrix}, 7e_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 7 \end{bmatrix}$$

Add them:

$$2e_0 + 5e_1 + 3e_2 + 7e_3 = \begin{bmatrix} 2 \\ 5 \\ 3 \\ 7 \end{bmatrix} = x$$

$$2e_0 + 5e_1 + 3e_2 + 7e_3$$
$$= x[0]e_0 + x[1]e_1 + x[2]e_2 + x[3]e_3$$
$$= \sum_{k=0}^{3} x[k] e_k$$

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Output for Any Signal

$$x = \sum_{k=0}^{N-1} x\,[k]e_k$$

We assume:

$$H(ax + by) = aH(x) + bH(y)$$

Apply H:

$$H(x) = H\left(\sum_{k=0}^{N-1} x\,[k]e_k\right)$$

Using linearity:

$$H(x) = \sum_{k=0}^{N-1} x\,[k]H(e_k)$$

This step is simply using linearity.

Substitute:

$$H(e_k) = S^k h$$

So

$$H(x) = \sum x\,[k]S^k h$$

This is convolution.

Fundamental Theorem

Every linear shift-invariant system is convolution.

$$y[n] = \sum_k x[k]\,h\,[n-k]$$

This is the fundamental formula of digital filtering.

Convolution as Matrix

Convolution Formula

$$y[n] = \sum_k x[k]h[n-k]$$

Example

$$h = (h_0, h_1, h_2)$$

Input (N=4)

$$x = (x_0, x_1, x_2, x_3)$$

Assume values outside range are **zero**:

$$x[-1] = x[-2] = 0$$

$$h[3] = h[4] = 0$$

This is called **zero padding**.

Compute y_0

Let $n = 0$

$$y_0 = \sum_k x[k]h[0-k]$$

Write possible terms

$$y_0 = x_0h_0 + x_1h_{-1} + x_2h_{-2} + x_3h_{-3}$$

But

$$h_{-1} = h_{-2} = h_{-3} = 0$$

So

$$y_0 = h_0x_0$$

Compute y_1

Now $n = 1$

$$y_1 = \sum_k x[k]h[1-k]$$

Expand

$$y_1 = x_0h_1 + x_1h_0 + x_2h_{-1} + x_3h_{-2}$$

Again

$$h_{-1} = h_{-2} = 0$$

So

$$y_1 = h_0x_1 + h_1x_0$$

Compute y_2

Let $n = 2$

$$y_2 = \sum_k x[k]h[2-k]$$

Expand

$$y_2 = x_0h_2 + x_1h_1 + x_2h_0 + x_3h_{-1}$$

Since

$$h_{-1} = 0$$

we get

$$y_2 = h_0x_2 + h_1x_1 + h_2x_0$$

Compute y_3

Let $n = 3$

$$y_3 = \sum_k x[k]h[3-k]$$

Expand

$$y_3 = x_0h_3 + x_1h_2 + x_2h_1 + x_3h_0$$

But the impulse response length is only 3

$$h = (h_0, h_1, h_2)$$

so

$$h_3 = 0$$

Therefore

$$y_3 = h_0x_3 + h_1x_2 + h_2x_1$$

Output Values

$$y_0 = h_0x_0$$

$$y_1 = h_0x_1 + h_1x_0$$

$$y_2 = h_0x_2 + h_1x_1 + h_2x_0$$

$$y_3 = h_0x_3 + h_1x_2 + h_2x_1$$

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Output Values

$$y_0 = h_0 x_0$$
$$y_1 = h_0 x_1 + h_1 x_0$$
$$y_2 = h_0 x_2 + h_1 x_1 + h_2 x_0$$
$$y_3 = h_0 x_3 + h_1 x_2 + h_2 x_1$$

Convert to Matrix Form

Group coefficients of x:

$$\begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} h_0 & 0 & 0 & 0 \\ h_1 & h_0 & 0 & 0 \\ h_2 & h_1 & h_0 & 0 \\ 0 & h_2 & h_1 & h_0 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$y = \begin{bmatrix} h_0 & 0 & 0 & 0 \\ h_1 & h_0 & 0 & 0 \\ h_2 & h_1 & h_0 & 0 \\ 0 & h_2 & h_1 & h_0 \end{bmatrix} x$$

This matrix is Toeplitz.

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Recall: Toeplitz matrix

Definition

Toeplitz matrix has constant diagonals.

Example (N=4):

$$T = \begin{bmatrix} h_0 & 0 & 0 & 0 \\ h_1 & h_0 & 0 & 0 \\ h_2 & h_1 & h_0 & 0 \\ h_3 & h_2 & h_1 & h_0 \end{bmatrix}$$

General rule:

$$T_{i,j} = h_{i-j}$$

Convolution to Matrix Form

Group coefficients of x:

$$\begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} h_0 & 0 & 0 & 0 \\ h_1 & h_0 & 0 & 0 \\ h_2 & h_1 & h_0 & 0 \\ 0 & h_2 & h_1 & h_0 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$y = \begin{bmatrix} h_0 & 0 & 0 & 0 \\ h_1 & h_0 & 0 & 0 \\ h_2 & h_1 & h_0 & 0 \\ 0 & h_2 & h_1 & h_0 \end{bmatrix} x$$

This matrix is Toeplitz.

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Applications and Importance





